

Computational Intelligence for Optimization - 2025/26

NOVA Information Management School

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**Approximating *Girl with a Pearl Earring*
Using a Genetic Algorithm with 100 Triangles**

Group C03_Galapagos

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1 Problem Description and Representation

This project uses a Genetic Algorithm (GA) to approximate Vermeer’s *Girl with a Pearl Earring* with exactly 100 solid-colour triangles. Quality is measured by the pixel-by-pixel distance to the 300×400 target image.

1.1 Individual Representation

- **Triangle geometry:** Each triangle’s coordinates is defined by three vertices: $\{(x_i, y_i)\}_{i=1}^3$.
- **Color encoding:** Each triangle is associated with four color parameters: red, green, and blue channel values in the range $[0, 255]$, together with an opacity value $\alpha \in [0, 1]$.
- **Chromosome layout.** A chromosome is represented as a list of 100 triangles. Each triangle is encoded using 10 genes: 6 values for geometry, 3 for RGB channels and 1 opacity value α . Consequently, each chromosome contains a total of 1000 genes.

2 Fitness Functions

2.1 Base Fitness: RMSE

The fitness function for an individual $\mathbf{x}$, selected by the professor for this problem, is defined as the Root Mean Squared Error (RMSE):

$$f_{\text{RMSE}}(\mathbf{x}) = \sqrt{\frac{1}{3WH} \sum_{i=1}^H \sum_{j=1}^W \sum_{c \in \{R,G,B\}} (\hat{I}_{ijc}(\mathbf{x}) - I_{ijc})^2} \quad (1)$$

where $\hat{I}_{ijc}(\mathbf{x})$ and I_{ijc} are the rendered and target image pixels at position (i, j) for channel $c \in \{R, G, B\}$, respectively; $W = 300$ and $H = 400$ represent the image width and height in pixels; and 3 is the number of color channels.

2.2 Alternative Fitnesses

2.2.1 Mean Squared Error (MSE)

Mean Squared Error (MSE) is the square of RMSE. Because the square root is strictly monotonic, minimizing MSE is equivalent to minimizing RMSE for rank-based selection. Omitting the square root makes MSE computationally faster for large-scale evaluations, though it sacrifices direct pixel-scale interpretability. However, algorithm interactions differ: while rank-based methods treat them identically, fitness-proportionate would have more selection pressure using MSE due to the exaggerated scaling of errors.

2.2.2 CIELAB / Delta E

Unlike standard RGB, the CIELAB color model [1] [2] was used to compare images because it more accurately mimics human vision. It evaluates differences by separating brightness from color information, rather than simply comparing raw pixel values.

$$f_{\Delta E_{ab}}(\mathbf{x}) = \frac{1}{WH} \sum_{i=1}^H \sum_{j=1}^W \sqrt{(\hat{L}_{ij} - L_{ij})^2 + (\hat{a}_{ij} - a_{ij})^2 + (\hat{b}_{ij} - b_{ij})^2} \quad (2)$$

where $(\hat{L}_{ij}, \hat{a}_{ij}, \hat{b}_{ij})$ and (L_{ij}, a_{ij}, b_{ij}) are the CIELAB pixel values of the rendered and target images at position (i, j) , respectively; L represents lightness, a the green-red axis and b the blue-yellow axis; $W = 300$ and $H = 400$ are the image width and height in pixels; and ΔE_{ab} corresponds to the Euclidean colour distance in the CIELAB colour space.

2.2.3 Embedding Cosine Distance

Alternatively, an embedding-based fitness metric evaluates semantic similarity by computing the cosine distance between high-level features extracted from the rendered and target images using DINOv2 [3] (`facebook/dinov2-base`). Lower values indicate greater similarity, bounded by zero.

$$f_{\text{emb}}(\mathbf{x}) = 1 - \frac{\phi(\hat{I}(\mathbf{x})) \cdot \phi(I)}{\|\phi(\hat{I}(\mathbf{x}))\| \|\phi(I)\|} \quad (3)$$

where $\hat{I}(\mathbf{x})$ and I are the rendered and target images, respectively; $\phi(\cdot)$ represents the feature vector extracted by the pre-trained DINOv2 model; the numerator computes the dot product between the two embedding vectors; the denominator normalizes both vectors by their magnitudes; and $f_{\text{emb}}(\mathbf{x})$ is the resulting cosine distance minimized by the genetic algorithm.

3 Genetic Algorithm Design

3.1 Initialisation

The initial population is randomly generated with each of the 10 genes per triangle sampled uniformly within their respective domains: coordinates within canvas bounds (width W and height H), RGB colours in $[0, 255]$, and alpha transparency in $[0, 1]$. We employed a population size of $\mu = 100$ individuals throughout training.

3.2 Selection Mechanisms

Three selection strategies were evaluated based on their evolutionary trade-offs. Fitness-proportionate selection is highly susceptible to premature convergence if a few high-fitness individuals dominate the population. Ranking selection mitigates this issue by enforcing smoother selection pressure, which better preserves population diversity. Finally, tournament selection offers a computationally simple and scalable alternative that allows for easily adjustable selection pressure [4].

3.3 Crossover Operators

Four crossover strategies were implemented and tested [5]:

- **Single-point crossover:** A random crossover point splits both parents; offspring inherit the left segment from one parent and right from the other.
- **Uniform crossover:** At each gene position, offspring inherits the entire triangle from one parent with probability 0.5 and from the other with probability 0.5; allows fine-grained mixing without positional bias.
- **Arithmetic crossover:** Matching triangles are blended using weighted interpolation with random weight $w \in [0, 1]$; enables smooth exploration in the continuous space of triangle configurations.
- **Order-based crossover:** A random subset of triangle positions is preserved from one parent; the remaining positions are filled preserving the relative order from the other parent.

3.4 Mutation Operators

Four mutation strategies were [5]:

- **Gaussian mutation:** Each gene (coordinates, colours, alpha) is independently perturbed by $\mathcal{N}(0, \sigma^2)$ with probability $p_m = 0.05$. Perturbations are applied with $\sigma_{\text{coords}} = 8.0$, $\sigma_{\text{color}} = 12.0$, $\sigma_{\text{alpha}} = 0.05$. Enables fine-tuned local search.
- **Triangle swap mutation:** Two randomly selected triangles exchange positions (Z-order); introduces structural variation in the layering sequence with probability $p_m = 0.05$.
- **Random reset mutation:** With probability $p_m = 0.01$, an entire triangle is replaced by a newly generated random triangle; allows escape from local optima via drastic restructuring.
- **Mixed mutation:** Randomly chooses between swap (structural, weight $w_s = 0.3$) and Gaussian (fine-tuning, weight $1 - w_s = 0.7$) mutation; combines local refinement with occasional structural jumps.

3.5 Diversity Preservation

Three diversity preservation strategies were implemented to counteract premature convergence:

- **Plateau Diversity:** If the best fitness does not improve by more than ε over a sliding window of w generations, the worst individuals are replaced by newly generated random solutions.
- **Fitness Sharing:** Each individual’s raw fitness is divided by the sum of a sharing function $\text{sh}(d_{ij})$ over all pairwise genotypic distances, reducing selection pressure on over-represented solutions [6].
- **Restricted Mating:** Crossover is restricted to pairs whose genotypic distance exceeds a minimum threshold, enforcing structural dissimilarity between parents.

4 Results

4.1 Random Search

To select the final configuration, a random hyperparameter search was performed over 48 trials, 200 generations, and 30 seeds. The Embedding Cosine Distance (DINOv2) was excluded from this phase, as passing nearly 28.8 million generated images through a deep neural network would be computationally prohibitive. The remaining metrics — RMSE, MSE, and CIELAB ΔE_{ab} — were compared to evaluate whether metric choice affects operator ranking.

One of the most important aspects to compare is whether the two fitness metrics — MSE and CIELAB — produce consistent conclusions across selection, crossover, and mutation operators, as disagreement between metrics would indicate that the choice of evaluation function affects the optimization outcome.

Below are the plots that show the median fitness over generation, grouped by operator type.

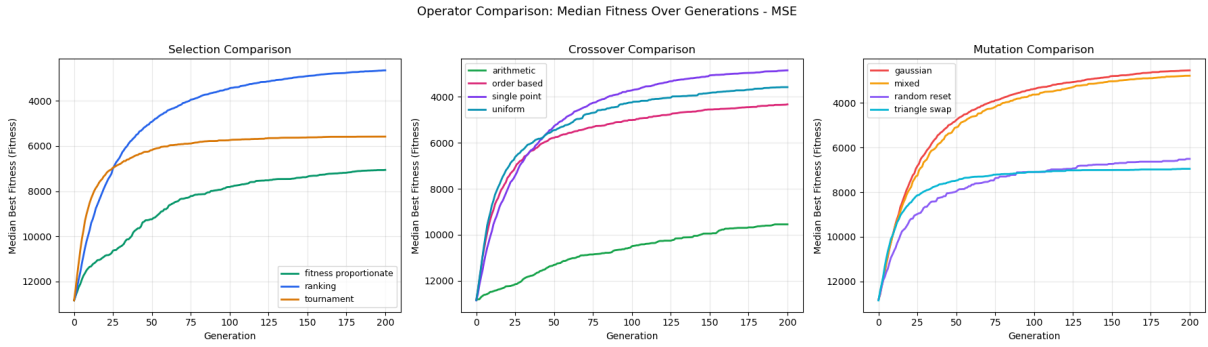


Figure 1: Operator comparison of median fitness over generations using MSE

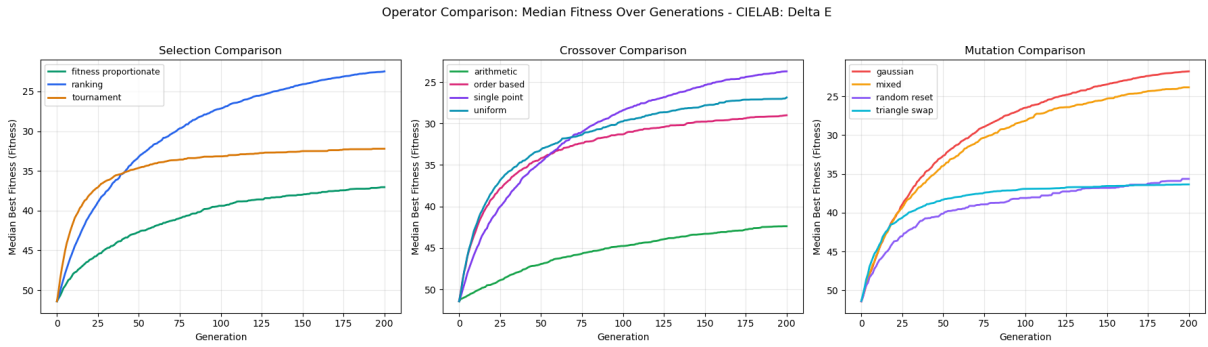


Figure 2: Operator comparison of median fitness over generations using CIELAB Delta E metric.

From these results, we could conclude many important characteristics.

- Regarding selection, both metrics consistently rank the operators in the same order: ranking selection performs best, followed by tournament, with fitness-proportionate selection in last place. However, as shown in Figures 4 and 5, ranking selection (despite proving more consistent and achieving stronger results under MSE) did not yield the best individual outcomes under CIELAB, where some trials reached a fitness almost as low as 15 after 200 generations, compared to MSE trials which barely dropped below 20.

- Arithmetic crossover proved itself to be the worst of the 4 techniques.
- Gaussian and Mixed performance behaved significantly better than Random Swap and Triangle Swap.

These results were further corroborated by analysing the top 20 configurations (top 10 by CIELAB Delta E and top 10 by MSE): fitness-proportionate selection was absent from all 20, as was random reset and triangle swap mutation. However, and to our surprise, an exception emerged with single-point crossover, which despite achieving the best median fitness under both metrics, it appears in only one of the top 10 MSE configurations and in none of the top 10 CIELAB configurations. To ensure the best configuration is genuinely superior, we performed a one-vs-all Wilcoxon signed-rank test comparing its final-generation fitness distribution against every other configuration.

4.2 Multiple Fitnesses

This section evaluates extended 6000-generation runs for each fitness function. To maximize performance, RMSE, MSE, and CIELAB ΔE were each run using the specific top-ranked hyperparameter configuration identified during their respective random searches. To ensure a fair baseline comparison, no hill-climbing refinement was applied to any run (Appendix B)

Note: Because the embedding metric was computationally excluded from the random search phase, its extended run utilized a custom configuration. The final quantitative results (Table 1) show that RMSE and MSE achieved the lowest pixel-level errors. The CIELAB ΔE approach yielded slightly higher errors across traditional metrics. As mathematically expected, the embedding method successfully minimized the cosine distance but heavily sacrificed spatial and pixel-level accuracy in the process.

Table 1: Final quantitative results for each optimized solution (no hill-climbing refinement applied).

Solution	RMSE	MSE	ΔE	Cosine distance
RMSE-optimized	28.180	794.088	11.493	0.8621
MSE-optimized	28.635	819.991	11.590	0.8860
ΔE -optimized	36.848	1357.798	12.402	0.9160
Embedding-cosine-optimized	138.828	19273.184	59.604	0.3132

5 RMSE Tuning

RMSE is the metric specified by the problem statement and was therefore used as the primary objective for all refinements. The best configuration from the random search (Tournament selection, Uniform crossover, Mixed mutation, $p_c = 0.78$, $p_m = 0.064$) was taken as the base and further improved through trial and error (parameters) and the two mechanisms described below.

5.1 Memetic Hill-Climbing

The GA was augmented with a hill-climbing (HC) procedure, making it a Memetic Algorithm. Every 300 generations, the best individual is locally refined: 15 neighbours are generated via Gaussian mutation ($p_m = 0.03$) for 80 iterations, and the fittest neighbour replaces the current solution if it improves fitness. Upon termination, a final intensive HC phase with reduced step sizes and more iterations is applied.

To validate this choice, a statistical comparison was run between the pure GA and the Memetic GA over 30 seeds and 1000 generations, with all other settings identical. The Memetic GA shows visible improvement steps precisely at the HC trigger points (Fig. 12), confirming that the local search is actively refining the best solution (Fig. 13). The Wilcoxon signed-rank [7] confirms the difference is statistically significant ($p < 0.001$) (Fig. 14): the Memetic GA consistently achieves lower final RMSE, justifying the additional computational cost.

5.2 Diversity Preservation

Three diversity strategies were compared against a clean baseline over 30 seeds and 200 generations (Fig. 15 and 16). Plateau Diversity and Fitness Sharing were statistically equivalent (Fig. 17) to the baseline ($p = 0.39$ and $p = 0.76$, respectively), while Restricted Mating was significantly worse ($p < 0.001$), as confirmed by the Wilcoxon signed-rank test [7]. Plateau Diversity was selected for the long run with a wider window ($w = 40$ generations) to avoid premature triggering.

6 Discussion and Conclusions

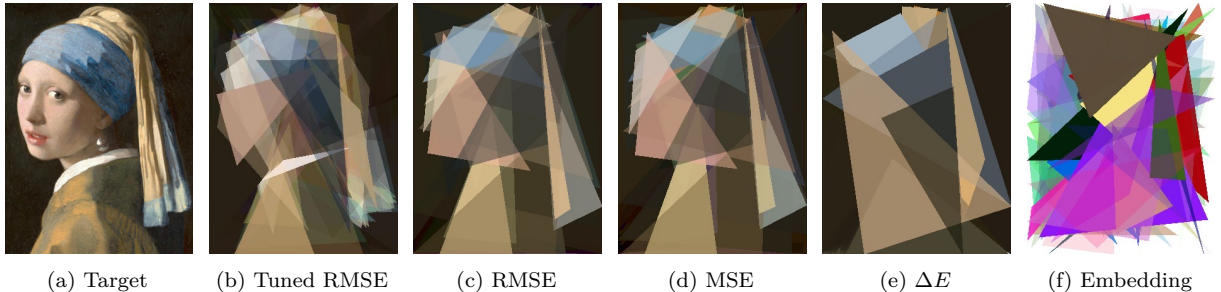


Figure 3: Visual comparison between the target image and the best results produced by each fitness function.

This project demonstrated that a Genetic Algorithm can approximate *Girl with a Pearl Earring* using a highly constrained representation (100 semi-transparent triangles), and that operator choices materially affect performance. Across trials, ranking/tournament selection and Gaussian/mixed mutation were consistently strong, while fitness-proportionate selection and arithmetic crossover lagged.

Fitness choice changes what “good” means: RMSE/MSE optimization achieved the best pixel-level accuracy (best RMSE ≈ 28.18), CIELAB ΔE produced competitive perceptual colour matching, and embedding-cosine favoured semantic similarity at the expense of spatial fidelity. Adding periodic hill-climbing significantly improved RMSE over the pure GA ($p < 0.001$), making the memetic variant the best overall approach within the project constraints.

References

- [1] K McLaren. “The development of the CIE 1976 ($L^*a^*b^*$) colour-space and colour-difference formula”. In: *Journal of the Society of Dyers and Colourists* 92.9 (1976), pp. 338–341.
- [2] Commission Internationale de l’Eclairage. *Colorimetry*. 3rd. Vienna, Austria: CIE Central Bureau, 2004.
- [3] Maxime Oquab et al. “DINOv2: Learning Robust Visual Features without Supervision”. In: *arXiv preprint arXiv:2304.07193* (2023).
- [4] David E. Goldberg and Kalyanmoy Deb. “A Comparative Analysis of Selection Schemes Used in Genetic Algorithms”. In: *Foundations of Genetic Algorithms*. Ed. by Gregory J. E. Rawlins. San Mateo, California: Morgan Kaufmann, 1991, pp. 69–93.
- [5] Thomas Bäck. *Evolutionary Algorithms in Theory and Practice*. New York: Oxford University Press, 1996.
- [6] Leonardo Vanneschi and Sara Silva. *Lectures on Intelligent Systems*. 1st. Natural Computing Series. Cham, Switzerland: Springer, 2023. ISBN: 978-3-031-17922-8. DOI: [10.1007/978-3-031-17922-8](https://doi.org/10.1007/978-3-031-17922-8).
- [7] D. Rey and M. Neuhäuser. “Wilcoxon-Signed-Rank Test”. In: *International Encyclopedia of Statistical Science*. Ed. by M. Lovric. Springer, 2011, pp. 1658–1659.

A Hyperparameter Search - Full Results

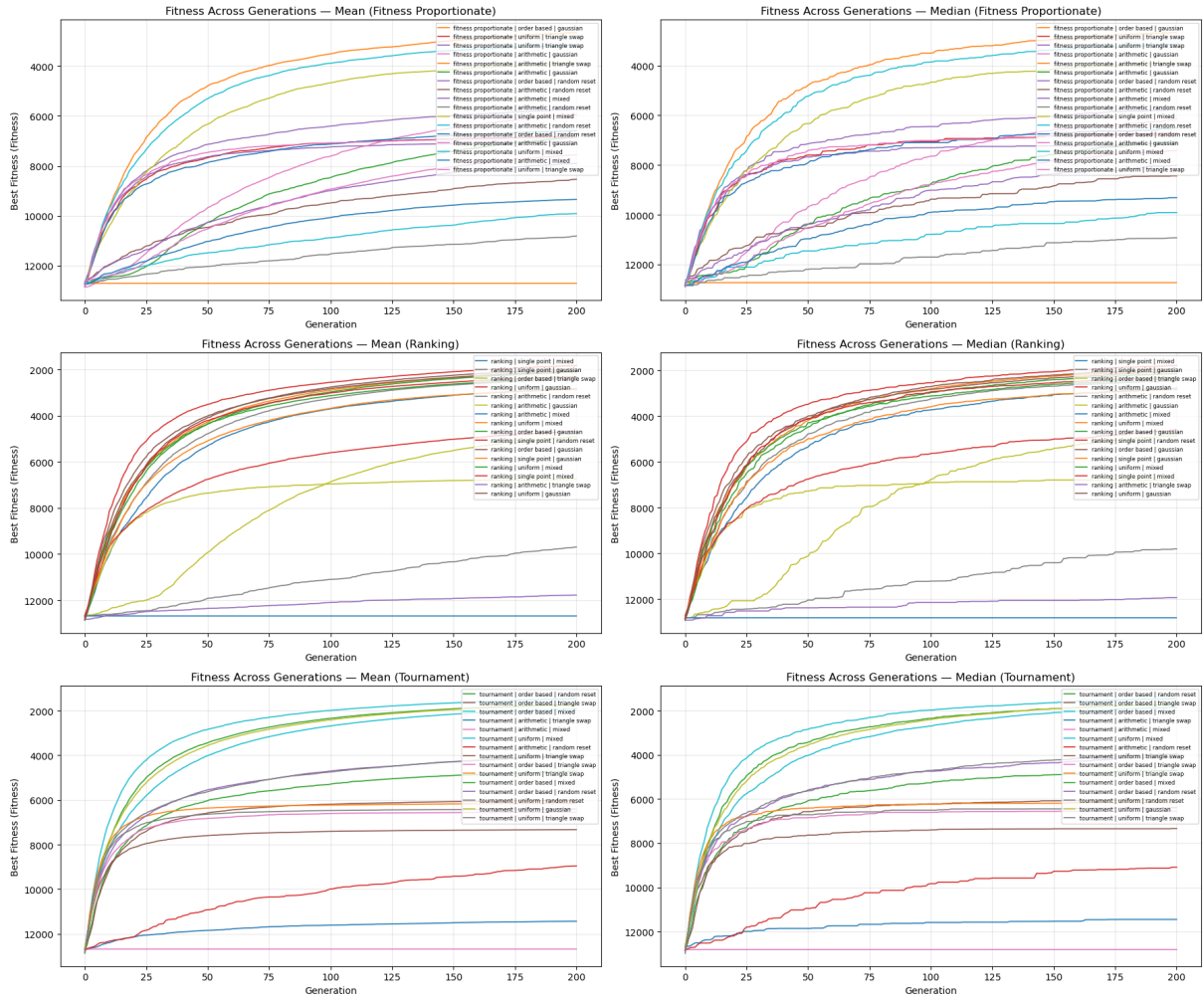


Figure 4: Mean (left column) and median (right column) best fitness per generation under the **MSE** metric, grouped by parameters

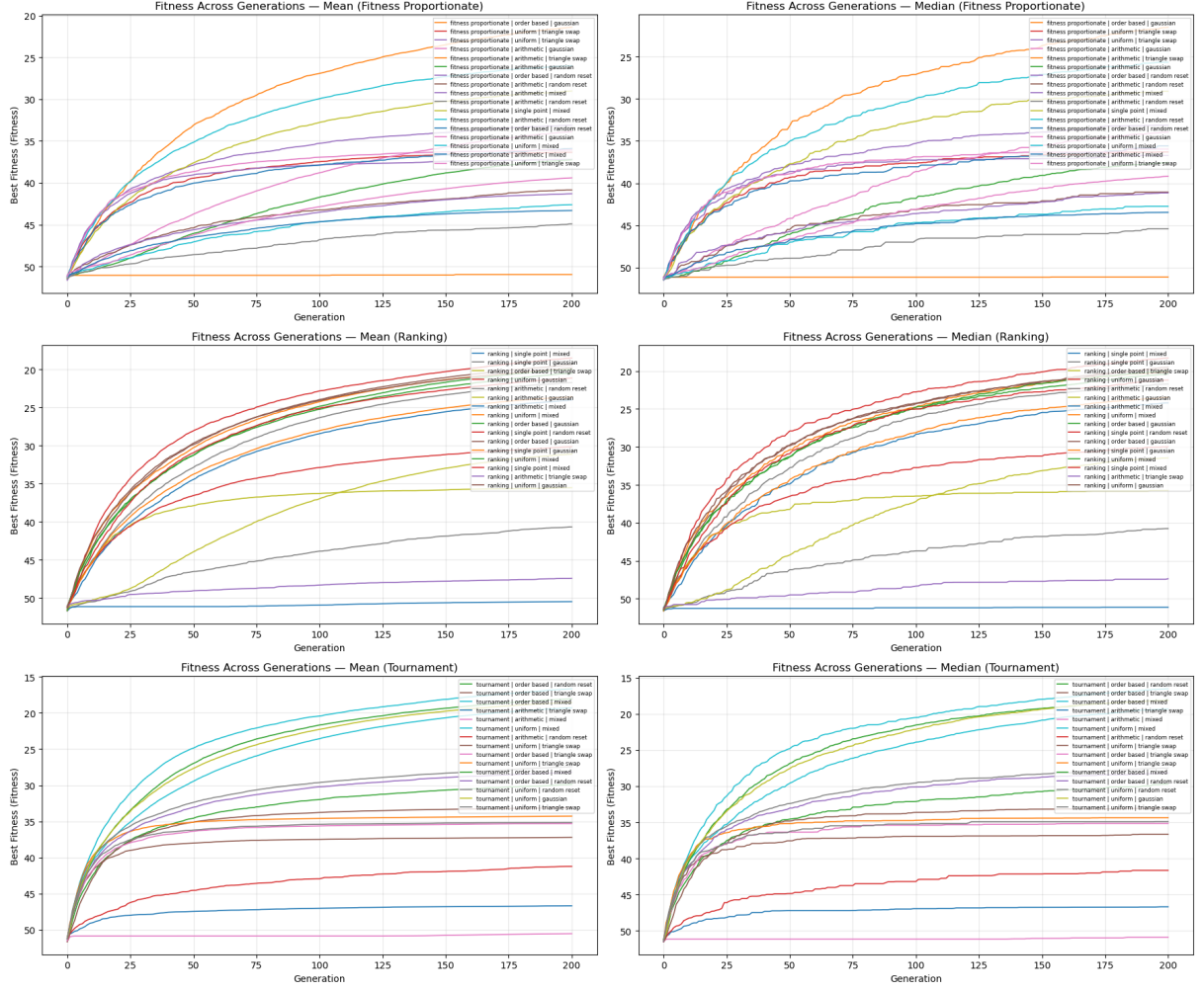


Figure 5: Mean (left column) and median (right column) best fitness per generation under the **CIELAB** metric, grouped by parameters.

B Multiple Fitnesses

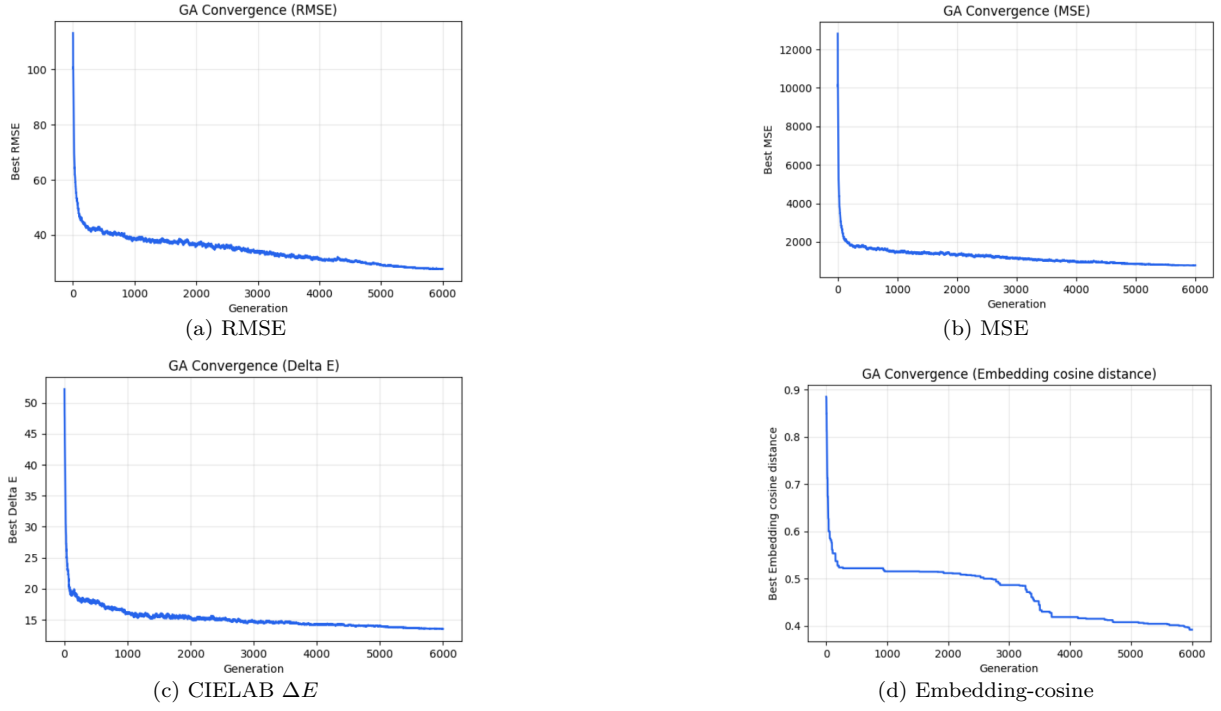


Figure 6: Operator comparison of the fitness over 6000 generations for RMSE, MSE, CIELAB ΔE , and embedding-cosine metrics.

C Timelapse Frames

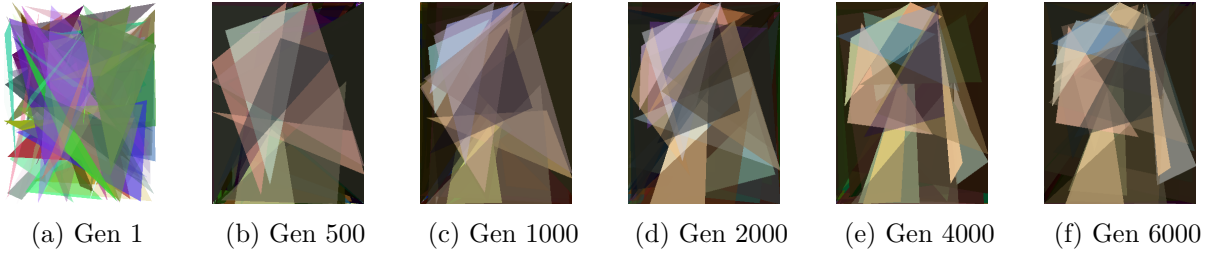


Figure 7: RMSE run — best individual at selected generations.

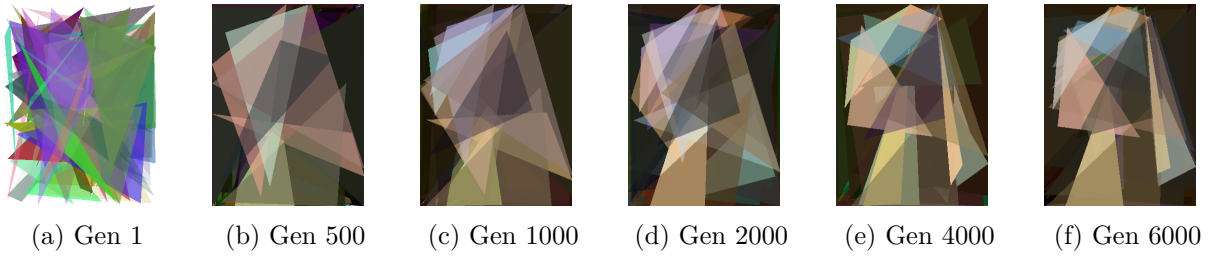


Figure 8: MSE run — best individual at selected generations.

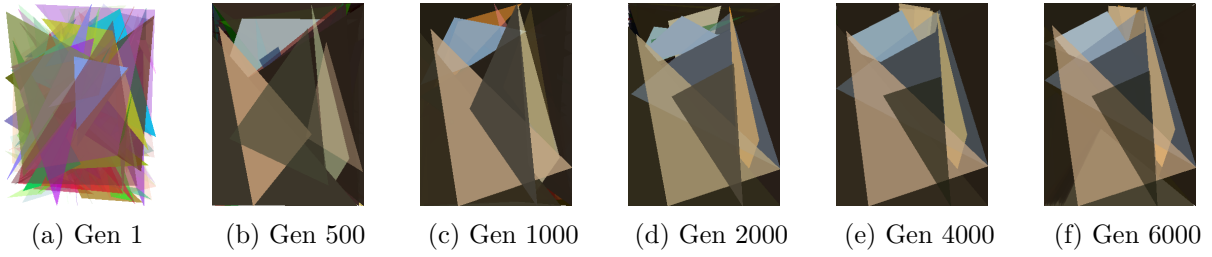


Figure 9: Cielab Delta E — best individual at selected generations.

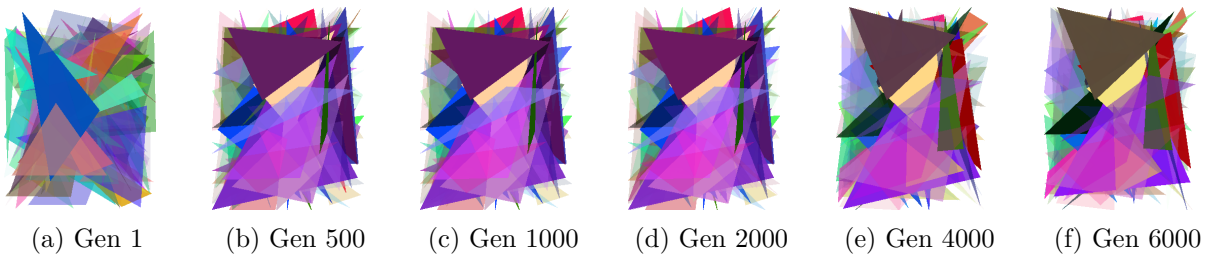


Figure 10: Embedding Cosine Distance run — best individual at selected generations.

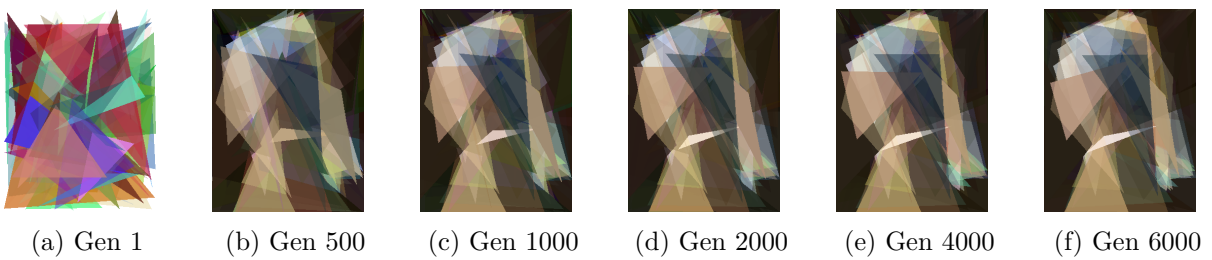


Figure 11: Tuned RMSE run — best individual at selected generations.

D Additional Content

Table 2: Top-5 configurations from random hyperparameter search.

Rank	Fitness	Pop	Selection	Crossover	Mutation	p_c	p_m
1	38.2012	100	Tournament	Uniform	Mixed	0.78	0.064
2	41.3431	100	Tournament	Order-based	Mixed	0.84	0.042
3	41.9248	100	Tournament	Uniform	Gaussian	0.85	0.015
4	42.8997	100	Ranking	Uniform	Gaussian	0.73	0.100
5	43.7973	100	Tournament	Order-based	Mixed	0.92	0.019

Table 3: Top 5 configurations found for the MSE fitness.

Rank	Fitness	Std.	Pop.	Selection	Crossover	Mutation	p_c	p_m
1	1462.0282	125.9825	100	Tournament	Uniform	Mixed	0.78	0.064
2	1712.3502	145.7708	100	Tournament	Order-based	Mixed	0.84	0.042
3	1761.5096	166.4148	100	Tournament	Uniform	Gaussian	0.85	0.015
4	1828.0858	156.9964	100	Ranking	Uniform	Gaussian	0.73	0.100
5	1924.5011	224.1306	100	Tournament	Order-based	Mixed	0.92	0.019

Table 4: Top 5 configurations found for the CIELAB ΔE_{ab} fitness.

Rank	Fitness	Std.	Pop.	Selection	Crossover	Mutation	p_c	p_m
1	16.6591	1.1868	100	Tournament	Uniform	Mixed	0.78	0.064
2	18.0518	0.9546	100	Tournament	Order-based	Mixed	0.84	0.042
3	18.4894	1.1605	100	Ranking	Uniform	Gaussian	0.73	0.100
4	18.5630	1.4593	100	Tournament	Uniform	Gaussian	0.85	0.015
5	19.2787	1.5685	100	Tournament	Order-based	Mixed	0.92	0.019

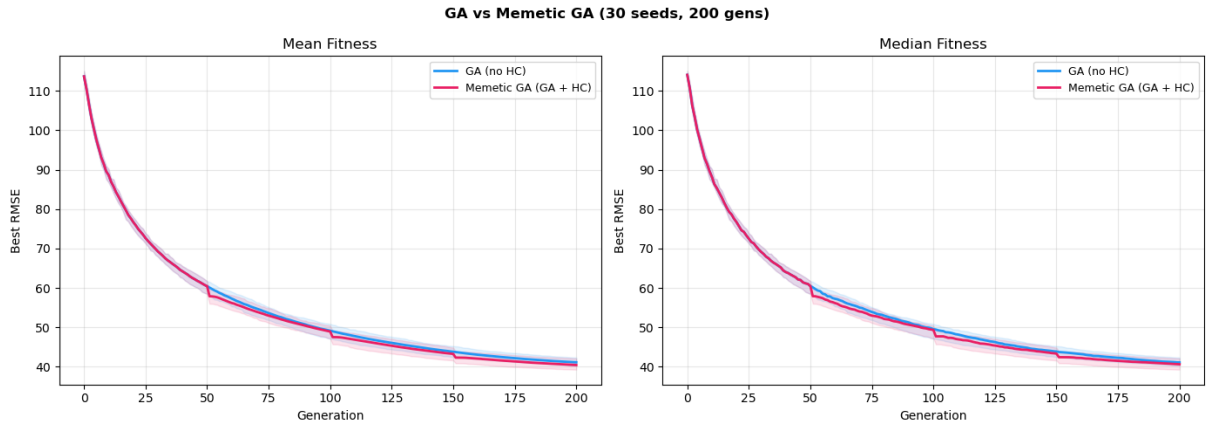


Figure 12: GA vs Memetic GA

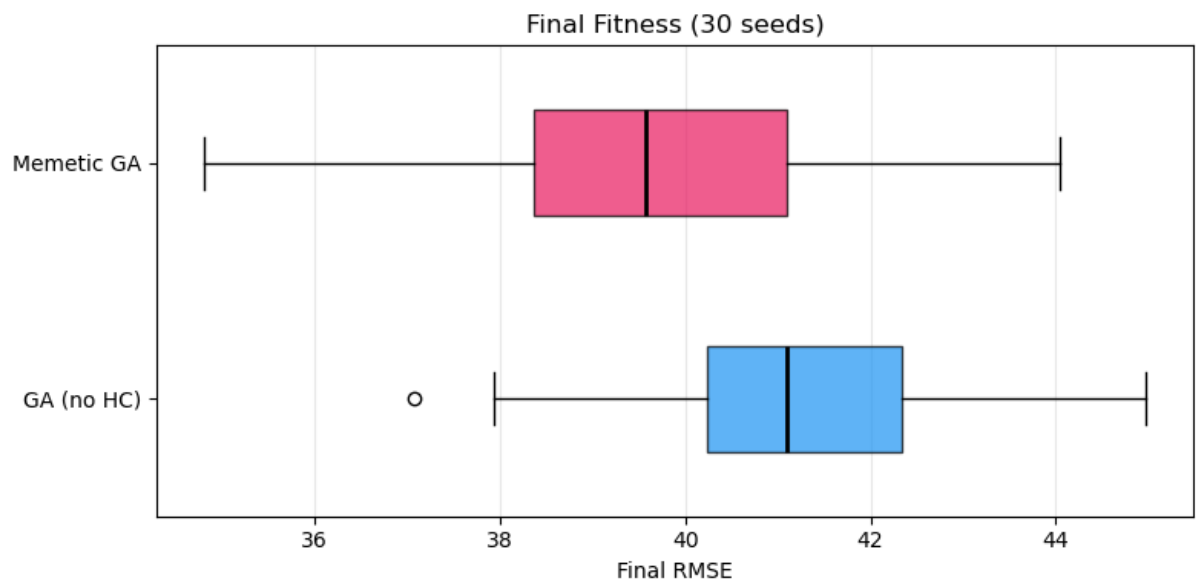


Figure 13: GA vs Memetic GA - Final RMSE

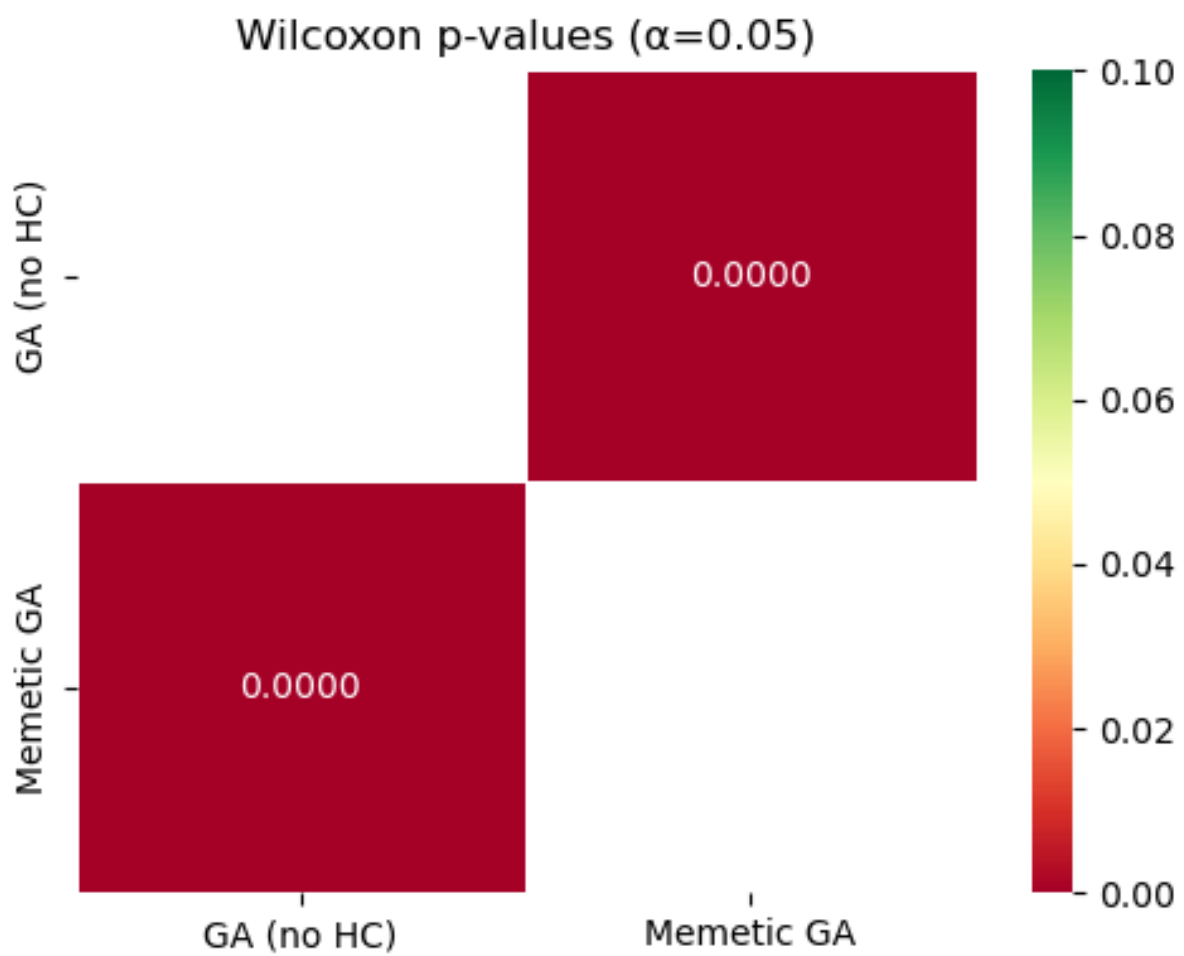


Figure 14: Wilcoxon p-values for GA vs Memetic GA

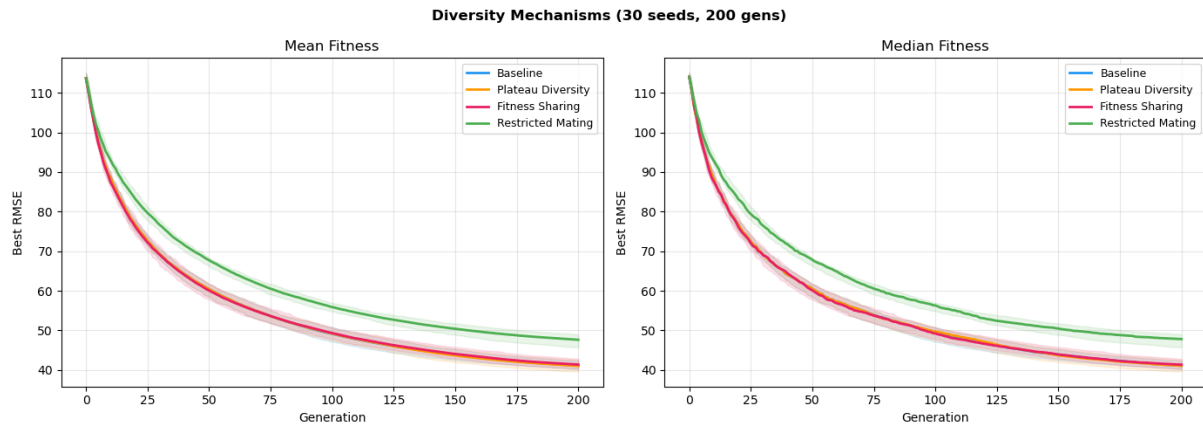


Figure 15: Diversity techniques test

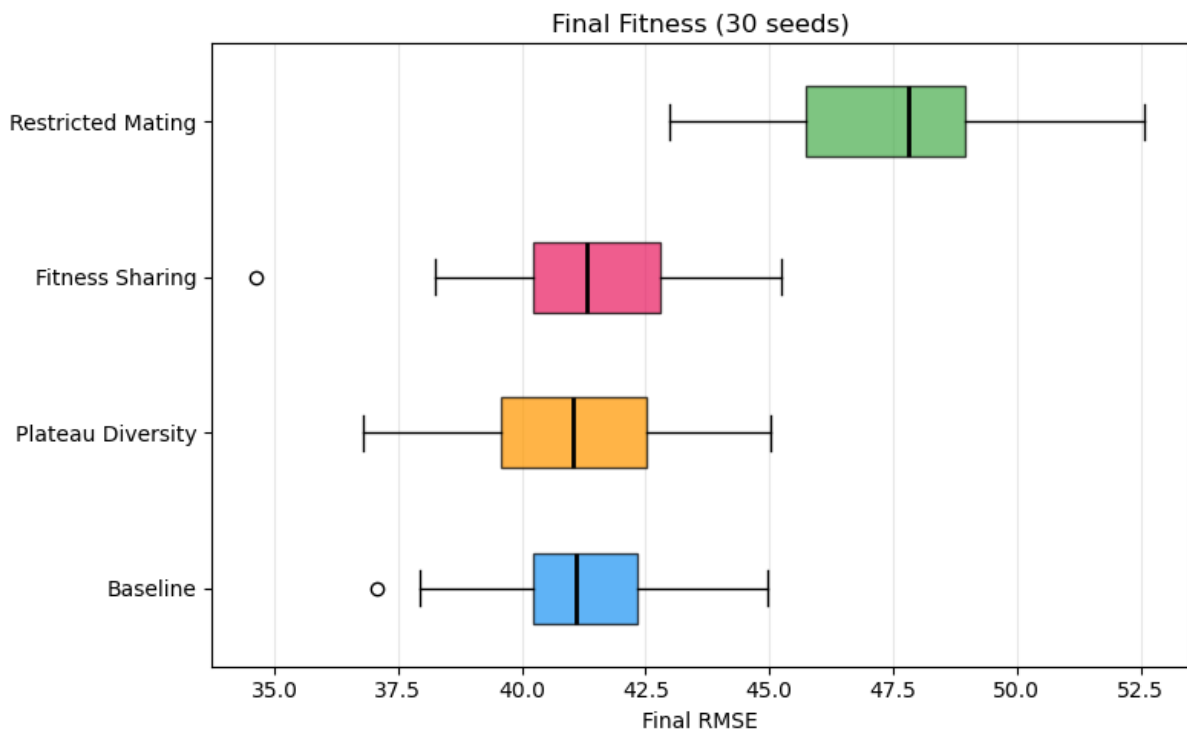


Figure 16: Diversity test - Final RMSE

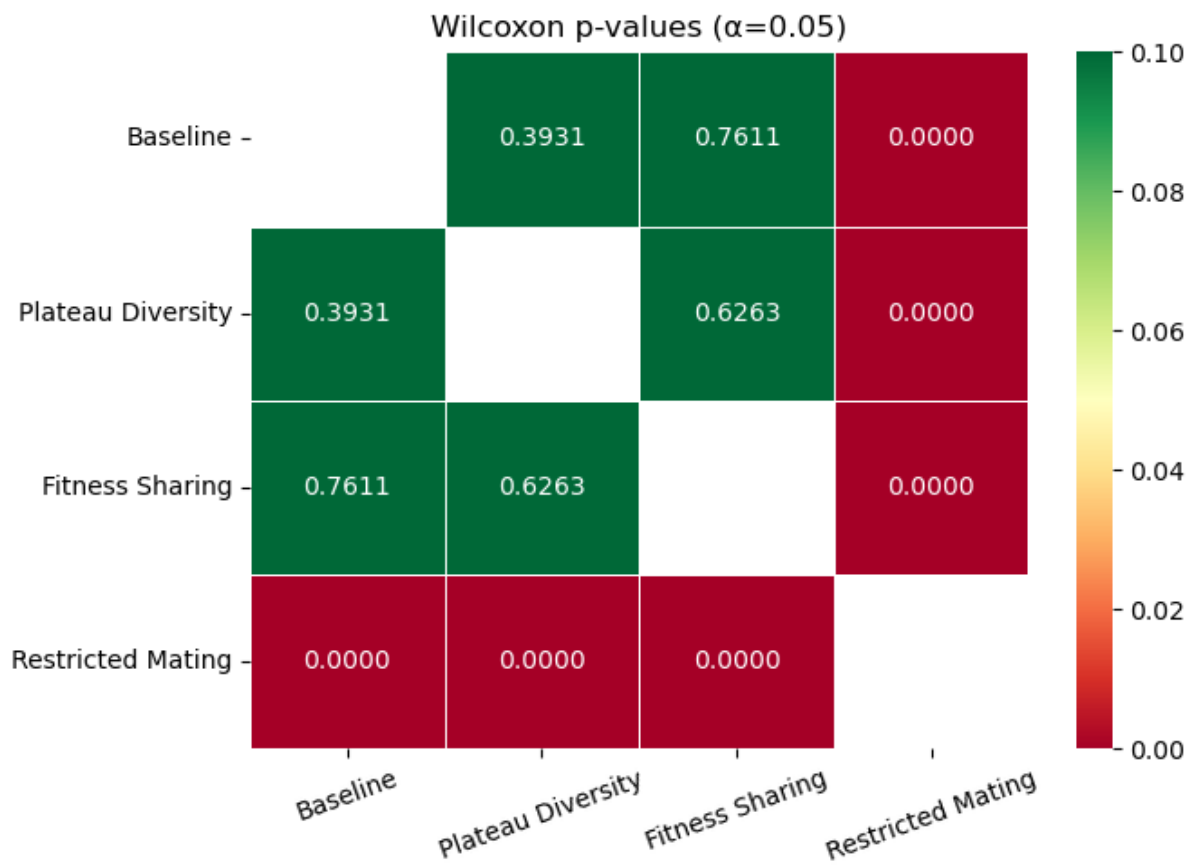


Figure 17: Diversity test p-value